

Computational modeling of UHF/SHF electromagnetic f intensity in enclosures with dielectric ma

Juan David Conde Duarte[1] | Faculty Advisor: Gladys Liliana Bautis

[1]Systems Engineering Program, Faculty of Engineering and Architecture, University of Pamplona, Pamplona, Colombia (ID 11023
[2]Course: Electromagnetism, Department of Physics, Faculty of Basic Sciences, University of Pamplona (gladys.bau

Abstract

The study of the spatial distribution of indoor electromagnetic waves is a fundamental problem in applied electromagnetism due phenomena caused by building structures. In this work, a two-dimensional computational model is developed to simulate stationary prop (2.4 GHz, $\lambda = 12.5\text{ cm}$) and SHF (5.0 GHz, $\lambda = 6.0\text{ cm}$) bands within enclosures conta domain is discretized using the centered finite-difference method, applying Dirichlet boundary conditions at radiating sources and coefficients based on the relative permittivity (ε_r) and conductivity (σ) of each material (drywall, brick, concrete, a solved via Successive Over-Relaxation (SOR), determining at each node the received power (dBm), the time-averaged Poynting mean-square electric field E_{rms} (V/m). Validation against the analytical log-distance and ITU-R P.1238 models yields square error of 1.84 dB, confirming the higher dielectric penetration of the 2.4 GHz band and the Faraday cage shielding

Keywords: *Electromagnetic Waves; Dielectric Attenuation; Poynting Vector; Finite Differences; Successive Over-Relaxation (SOR).* | **Modalit**

Introduction and motivation

When a microwave electromagnetic wave propagates inside an enclosed space, its energy flux density decreases due to both geometric wavefront divergence and dissipation within walls characterized by relative permittivity ε_r and effective conductivity $\sigma > 0$ [1, 2]. While closed-form free-space analytical equations are only valid in homogeneous media, numerical modeling enables solving realistic architectural geometries.

General objective: To numerically model and validate the spatial distribution of received power (P_{rx}), time-averaged Poynting vector magnitude ($\langle S \rangle$), and root-mean-square electric field intensity (E_{rms}) in the 2.4 GHz and 5.0 GHz bands inside 2D enclosures with dielectric and conductive materials using finite differences and SOR relaxation.

Theoretical framework / Physical model

In air (intrinsic impedance $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 376.7\,\Omega$), the time-averaged magnitude of the **Poynting vector** $\langle S \rangle$ (W/m^2) relates to the RMS electric field E_{rms} (V/m), the RMS magnetic field H_{rms} (A/m), and the received power P_{rx} captured by an isotropic effective aperture $A_{\text{eff}} = \lambda^2/(4\pi)$ as follows [1]: $$\langle S \rangle = \frac{E_{\text{rms}}^2}{\eta_0} = \frac{4\pi\,P_{\text{rx}}}{\lambda^2} \implies E_{\text{rms}} = \sqrt{\frac{4\pi\,\eta_0\,P_{\text{rx}}}{\lambda^2}}, \quad H_{\text{rms}} = \frac{E_{\text{rms}}}{\eta_0} \quad (1)$$ When traversing a lossy dielectric material or a conductor, the field attenuates exponentially according to the Beer-Lambert law $E(x) = E_0 e^{-\alpha x}$. In a 2D stationary regime, the normalized energy intensity envelope $u(x,y) \in [0, 100]$ over the domain Ω with spatial absorption $\kappa^2(x,y)$ is governed by the elliptic partial differential equation [3]: $$\nabla^2 u(x,y) - \kappa^2(x,y)\,u(x,y) = 0 \quad \text{in } \Omega \quad (2)$$ subject to fixed **Dirichlet** boundary conditions at the radiating source $u(x_0,y_0) = u_{\text{EIRP}}$ and null absorbing boundaries along the outer perimeter $u|_{\partial\Omega} = 0$.

Description of the modeled system

The developed software (*SimuNet Física*) discretizes $12\text{ m} \times 12\text{ m}$ enclosures into 24×24 node grids ($h = 0.5\text{ m}$), allowing placement of sources operating at 2.4 GHz ($\lambda = 12.5\text{ cm}$) and 5.0 GHz ($\lambda = 6.0\text{ cm}$) as well as walls with the constitutive properties listed in Table 1 [4].

Table 1. Electromagnetic properties and attenuation losses of modeled media (ITU-R P.2040).

Material / Medium	Permittivity ε_r	Conductivity σ (S/m)	Loss (dB)	Base Coeff. $\alpha_{i,j}$
Air (Free Space)	1.0006	0.000	0 dB	0.0012
Drywall (Plasterboard)	2.8	0.015	3 dB	0.0200
Brick Wall	4.5	0.038	8 dB	0.0800
Reinforced Concrete	6.5	0.085	15 dB	0.1500
Metal (Faraday Cage)	—	3.5×10^7	≥ 35 dB	0.2500

Computational methodology

- **5-Point Finite-Difference Discretization:** Approximating the Laplacian operator on the 2D grid with the local dielectric absorption factor $\alpha_{i,j}(f, \varepsilon_r, \sigma)$: $$u_{i,j}^{\text{GS}} = \big(1 - \alpha_{i,j}\big)\,\frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}}{4} \quad (3)$$

Results and discussion

Figure 1 presents the two-dimensiona electric field obtained numerically. It is v = 12.5\text{ cm}$) achieves deeper pe ($5.0\text{ GHz}$, $\lambda = 6.0\text{ absorption by an additional $8.5\text{ d the limiting **Faraday Cage** effect (skin 1.3\,\mu\text{m}$), nullifying the exterior

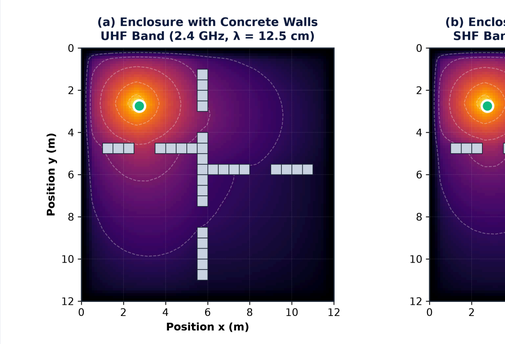


Figure 1. 2D finite-difference simulation: (a) (c) metalli

In Figure 2(a), the numerical radial pr model, achieving a root-mean-square 0.978$. Figure 2(b) and Table 2 demonstr required iterations by **73.7%** compared t

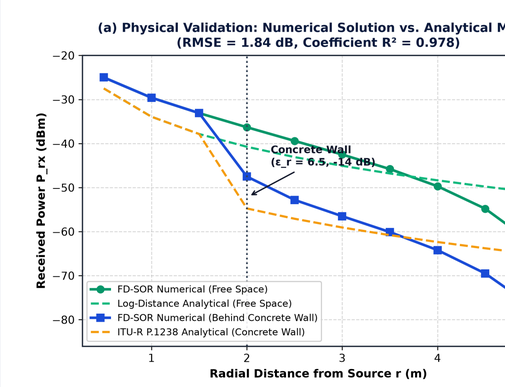


Figure 2. (a) Radial profile validation against Jacobi,

Table 2. Performance of numerical methods

Method / Scenario at 3 m	Iterations	Time	$P_{\t (
Jacobi ($\omega = 1.00$, simultaneous)	251	11.4 ms	
Gauss-Seidel ($\omega = 1.00$, successive)	138	6.2 ms	
Optimal SOR ($\omega = 1.45$)	**66 (-73.7%)**	**2.8 ms**	
Free Space (2.4 GHz, no walls)	66	2.8 ms	
Faraday Cage Exterior ($\sigma = 3.5\times	66	2.8 ms	